

Predicting and Explaining Traffic Collision Severity in Toronto Using Explainable Machine Learning

Problem Definition | Dataset Description | Research Questions

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1. Problem Definition

1.1 Background

Traffic collisions in Toronto continue to result in significant injuries, fatalities, and economic costs each year. While the City of Toronto provides detailed collision data through its Open Data Portal, most existing analyses remain descriptive — they summarize what has happened but do not offer predictive insights into what is likely to happen under specific conditions.

Collision outcomes are influenced by a complex combination of factors such as time of day, weather, road surface, location, and the type of road users involved. Traditional statistical summaries fail to capture these non-linear relationships. Furthermore, even when machine learning models are applied to accident data, they are often treated as black boxes — they produce a prediction but cannot explain why a collision was classified as severe. This limits their practical usefulness for policy and safety planning.

1.2 The Problem Statement

There is a clear need for a data-driven and explainable machine learning framework that can:

1. Accurately predict whether a traffic collision will result in a minor injury, major injury, or fatality, based on temporal, environmental, and spatial features.
2. Identify which specific factors most strongly contribute to severe collision outcomes.
3. Provide clear, interpretable explanations of model predictions using Explainable AI (XAI) techniques, so that results are useful for city planners, traffic engineers, and policy-makers.
4. Identify high-risk geographic zones across Toronto to support targeted road safety interventions.

1.3 Why This Problem Matters

The social and economic cost of traffic collisions in Toronto is substantial. Severe collisions place enormous pressure on emergency services, hospitals, and legal systems, and cause lasting harm to individuals and families. Road traffic crashes cost Canadian society billions of dollars annually in medical care, legal proceedings, vehicle damage, and lost productivity.

This research is directly aligned with Toronto's Vision Zero Road Safety Plan, which aims to eliminate all traffic-related deaths and serious injuries. A predictive and explainable model of the kind proposed here would give transportation planners an evidence-based tool to prioritize interventions and measure their effectiveness over time.

2. Dataset Description

This project uses two publicly available datasets that are merged to form one comprehensive dataset for training and evaluating the machine learning models.

2.1 Primary Dataset: City of Toronto Traffic Collision Open Data

Source: City of Toronto Open Data Portal — <https://open.toronto.ca/catalogue/>

The primary dataset contains officially recorded traffic collision events reported to the Toronto Police Service. Each row represents one collision incident. The dataset spans multiple years and contains approximately 60,000 to 80,000 records with around 30 structured variables.

Key variables include:

Variable	Type	Description
OCC_DATE	Date	Date of the collision occurrence
OCC_HOUR	Numerical	Hour of day (0–23) when collision occurred
OCC_MONTH	Categorical	Month of the collision
OCC_YEAR	Numerical	Year of the collision
OCC_DOW	Categorical	Day of week (Monday through Sunday)
LAT_WGS84 / LONG_WGS84	Numerical	GPS coordinates for spatial mapping
FATALITIES	Numerical	Count of fatalities per collision (target component)
INJURY_COLLISIONS	Binary	Flag: any injury occurred
PD_COLLISIONS	Binary	Flag: property damage only, no injury/fatality
NEIGHBOURHOOD_158	Categorical	Toronto neighbourhood (one of 158 zones)
DIVISION	Categorical	Toronto Police Service reporting division
AUTOMOBILE / BICYCLE / PEDESTRIAN	Binary	Flags for road user type involved in collision

Table 1: Key variables in the City of Toronto Traffic Collision Dataset

The target variable (collision severity) will be derived from the combination of FATALITIES, INJURY_COLLISIONS, and PD_COLLISIONS into three labelled classes: Minor (property damage only — class 0), Major (one or more injuries — class 1), and Fatal (one or more fatalities — class 2).

2.2 Secondary Dataset: Government of Canada Historical Climate Data

Source: Environment and Climate Change Canada — <https://climate.weather.gc.ca/>

Hourly weather data from the Toronto Pearson International Airport weather station will be merged with the collision dataset using the date and hour of each collision as the join key. This allows environmental conditions at the exact time of each collision to be included as model features.

Environmental variables to be included:

- Temperature (°C): Ambient temperature at the time of collision
- Total Precipitation (mm): Combined rain and melted snow, measured hourly
- Snowfall (cm): Snow accumulation in the hour prior to collision
- Wind Speed (km/h): Speed of wind at time of collision
- Visibility (km): Reported horizontal visibility distance
- Weather Condition Description: Categorical label (Clear, Fog, Rain, Snow, Blowing Snow, Freezing Rain, etc.)

2.3 Combined Dataset

After merging the two datasets on date and hour, the combined dataset will contain approximately 60,000 to 80,000 records with over 40 features covering temporal, spatial, environmental, and incident-specific variables. Both datasets are publicly available, fully anonymized, and free of personally identifiable information (PII), making them ethically appropriate for research use without requiring special data access agreements.

3. Research Questions

The following six research questions guide the direction of this project. Each question is connected to a specific phase of the analysis and will be addressed through the modeling and interpretation stages.

RQ1: How accurately can machine learning models predict traffic collision severity in Toronto?

This is the primary predictive question of the project. Using evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, the study will assess how well each model classifies collisions into the three severity categories (minor, major, fatal). Particular attention will be given to the recall of major and fatal classes, since correctly identifying the most dangerous collisions is the primary safety goal.

RQ2: Which temporal, environmental, and location-based factors most strongly influence severe collision outcomes?

Using SHAP values and feature importance analysis, the study will determine which variables — such as time of day, weather conditions, neighbourhood, and season — have the greatest impact on predicted collision severity. The goal is to produce a ranked list of risk factors that is both statistically grounded and practically interpretable.

RQ3: Do ensemble methods outperform traditional statistical models for collision severity prediction?

Logistic Regression will serve as the baseline model. This question asks whether ensemble methods — Random Forest, XGBoost, and Gradient Boosting — achieve meaningfully better predictive performance. The comparison will use identical training and test splits and the same evaluation metrics across all models to ensure a fair comparison.

RQ4: Can explainable AI techniques provide clear and actionable insights into the key drivers of severe collisions?

Beyond prediction accuracy, this project asks whether SHAP explanations are meaningful and useful for non-technical decision-makers. The study will evaluate whether SHAP output aligns with domain knowledge (e.g., known risk factors such as poor lighting and adverse weather) and whether it reveals any previously unknown interaction effects that could inform safety planning.

RQ5: Are certain neighbourhoods, time periods, or environmental conditions associated with higher predicted severity risk?

This spatial and temporal subgroup analysis will determine whether collision risk is concentrated in specific parts of Toronto or at specific times. Using geospatial mapping and DBSCAN clustering, the study will identify high-risk zones and evaluate whether model predictions are consistent with observed severity patterns in those zones.

RQ6: Does model performance vary across geographic regions or seasonal conditions within Toronto?

A model that performs well overall may still produce poor predictions for specific subgroups. This question asks whether performance degrades for certain neighbourhoods (e.g., suburban vs. downtown) or seasons (e.g., winter vs. summer). Subgroup evaluation ensures the model is robust and reliable across the diverse conditions present in a large Canadian city.